Baseline model evaluation

YOLOv8s fine-tuned to detect four hand gestures · Ben Nichiporik, Daffodil Ho, Joseph Lee, Thu Nguyen · November 2026

0.990
mAP50, validation (HaGRID, 160 images)

0.845
mAP50, our own photos (128 images, headline)

0.64
fist recall on our photos (weakest class, conf 0.25)

16.6 ms
per image on an RTX 5060 Ti (≈60 frames/s)

The baseline is close to perfect on held-out HaGRID images but loses 0.15 mAP50 on the photos our group collected. Its weakest spots are **small hands in wide shots** (only 56% found) and **fists** (64% found), which the model either misses or confuses with a thumbs-down. The model is fast enough for a live photo booth, and a higher confidence threshold (0.55) trades a little recall for noticeably fewer false triggers.

1. What was evaluated

Model. YOLOv8s pretrained on COCO, fine-tuned on our data (transfer learning; all layers trained). 50 epochs were run; the checkpoint with the best validation fitness (epoch 41) is the baseline, `models/baseline_best.pt`.

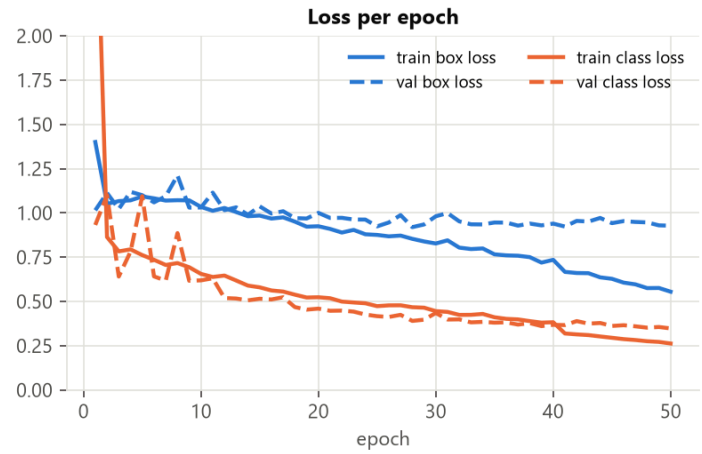
Protocol. Every model in the project is scored by the same code (`gesture_eval.py`) on three sets: HaGRID validation (used to pick the checkpoint), the full test split, and **our own photos only**, the honest estimate of photo-booth performance. Sections 4-6 then score our photos at a deployment threshold (confidence ≥ 0.25, a prediction counts as correct if its box overlaps a labelled hand with IoU ≥ 0.5 and the class matches).

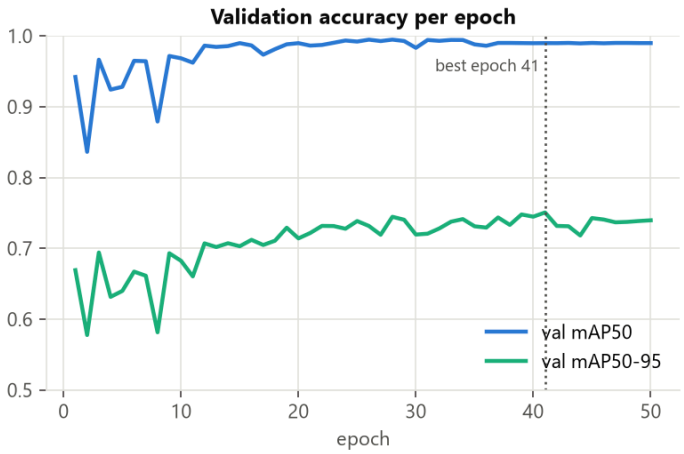
image size	640	**optimizer**	SGD
epochs	50 (patience 15)	**lr0**	0.01
batch	16	**momentum**	0.937
seed	0 (deterministic)	**weight decay**	0.0005
flips	LR 0.5, UD 0.0	**other aug.**	Ultralytics defaults (mosaic, HSV, scale)

split · source	palm	fist	like	dislike	total
train · HaGRID	320	320	320	320	1280
val · HaGRID	40	40	40	40	160
test · HaGRID	8	8	8	8	32
test · our photos	32	32	32	32	128

Images per split and class. HaGRID images 001-368 per class; our photos are 369-400 (8 per member per class).

2. Training behaviour





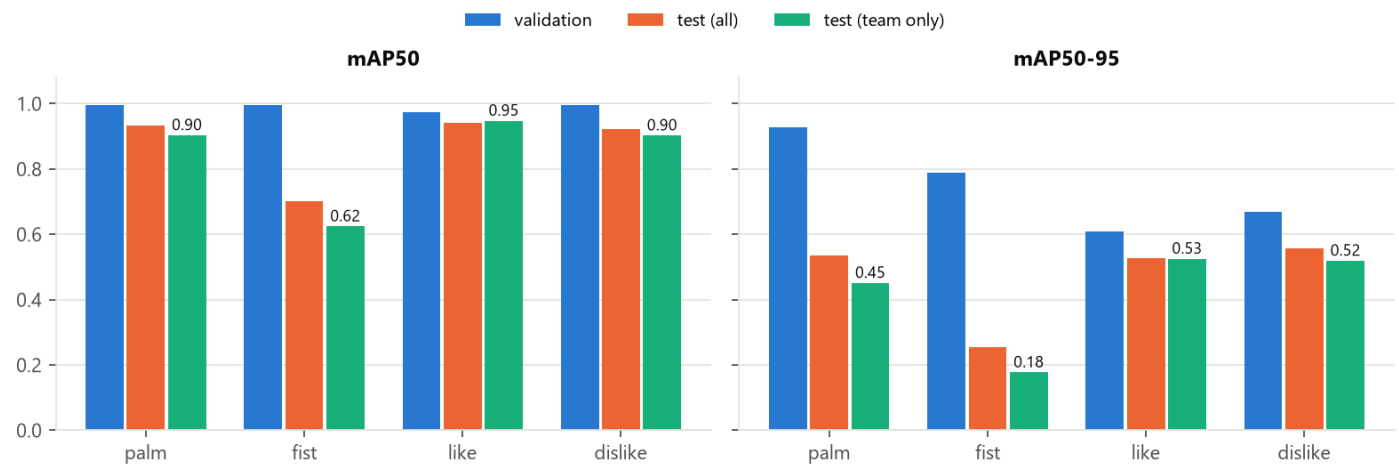
Transfer learning works quickly: validation mAP50 is already 0.94 after one epoch, because the COCO-pretrained backbone already provides general shape and edge features. Accuracy then climbs slowly. The best epoch (41) reaches mAP50 0.990 and mAP50-95 0.751. Validation losses keep falling alongside the training losses through epoch 50, so there is no sign of overfitting *to HaGRID*. Early stopping never triggered. The overfitting that does matter is to HaGRID's style of photo, which only the test on our own photos reveals.

3. Overall results

evaluation set	precision	recall	F1	mAP50	mAP50-95
validation (160)	0.990	0.984	0.987	0.990	0.748
test, all (160)	0.911	0.811	0.858	0.875	0.469
test, our photos only (128)	0.894	0.775	0.830	0.845	0.418

The validation scores are inflated: HaGRID validation images come from the same collection as the training images (one person standing and facing a phone camera, mostly indoors, one hand per photo). On our own photos, mAP50 drops by 0.15 and mAP50-95 by 0.33. The larger drop in mAP50-95 means the model usually *finds* the hand but its box doesn't line up tightly with ours. Correct detections have a mean IoU of only 0.76 with our labels. Part of this is annotation style rather than model error: each member drew boxes in MakeSense slightly differently (some include the wrist). At conf 0.25 on our photos, the 95% bootstrap intervals (resampling photos) are precision 0.77-0.90, recall 0.74-0.87 and F1 0.76-0.88. A single model's F1 is uncertain by about ±0.06, so small differences between tuned models should not be over-read.

4. Per-class performance

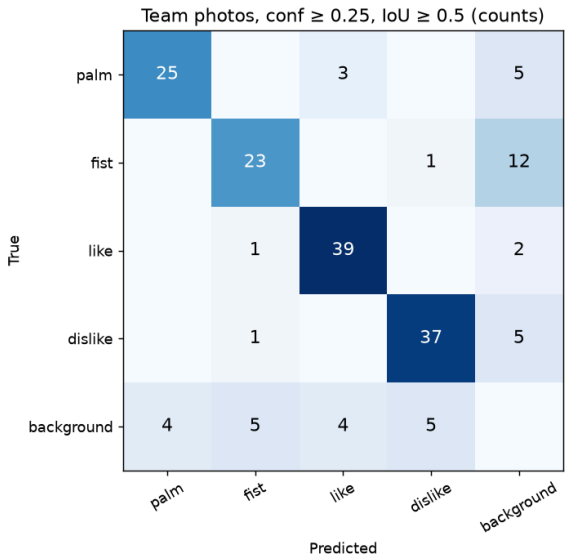


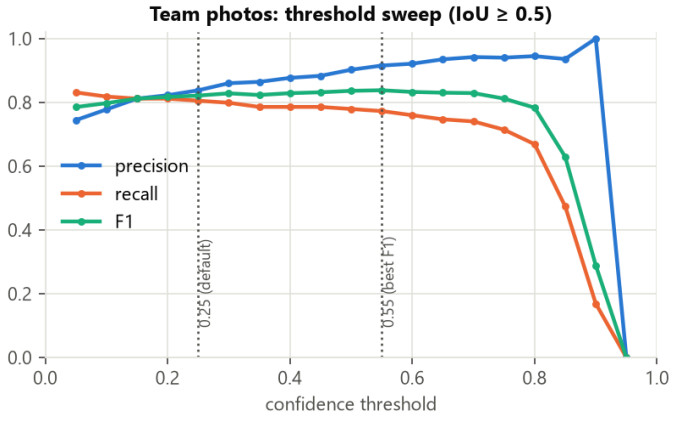
our photos	precision	recall	mAP50	mAP50-95	hands	missed @0.25	wrong class @0.25
palm	0.927	0.765	0.904	0.452	33	5	3
fist	0.758	0.611	0.625	0.177	36	12	1
like	0.927	0.909	0.948	0.526	42	2	1
dislike	0.963	0.814	0.903	0.519	43	5	1

Precision, recall and mAP from Ultralytics; missed and wrong-class counts at conf 0.25, IoU 0.5.

Palm and like transfer well: their shapes are distinctive (spread fingers, upward thumb). **Fist is the weakest class**: 12 of 36 fists are missed entirely (3 of them only because their labels were drawn too loosely; Section 8). A fist is a compact, low-texture blob that looks like many other things at small scale, and several of our fists are raised in crowd scenes with busy backgrounds, which HaGRID never shows (Section 6). **Dislike** is confused with fist in both directions: a thumbs-down seen from the side is a fist with a short thumb, and these are the highest-confidence errors (Section 6).

5. Errors at the deployment threshold





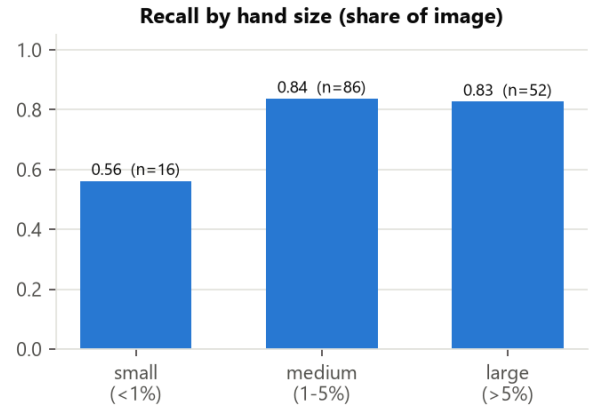
At conf 0.25 the model makes 148 predictions for 154 labelled hands in our 128 photos: **124 correct**, **24 hands missed**, **6 found with the wrong class**, and **18 predictions on background** (no labelled hand there; by predicted class: dislike 5, fist 5, palm 4, like 4). Missing a hand is the most common error. Ultralytics' own confusion matrix uses a near-zero

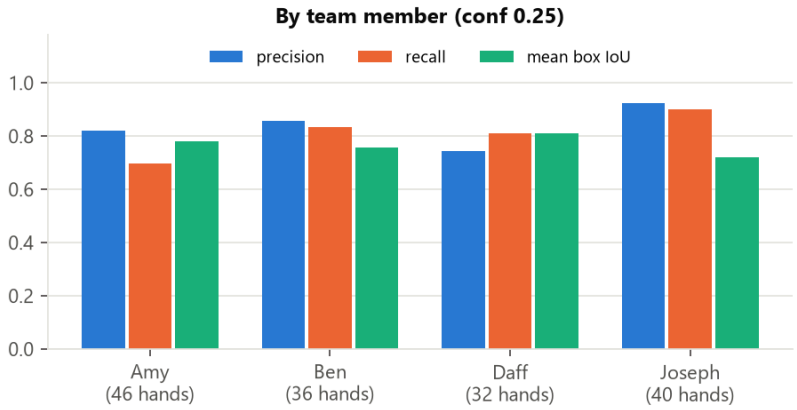
confidence threshold, so its background row looks far worse than what a user would see. The matrix here uses the real threshold.

Choosing the threshold. F1 is highest at conf 0.55 (precision 0.92, recall 0.77, F1 0.84), compared with precision 0.84 and recall 0.81 at 0.25. For a photo booth a false trigger (taking a picture nobody asked for) is worse than a missed gesture (the user just holds it a moment longer), so we recommend conf ≈ 0.55 plus requiring the gesture over several consecutive frames.

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6. Where the model fails

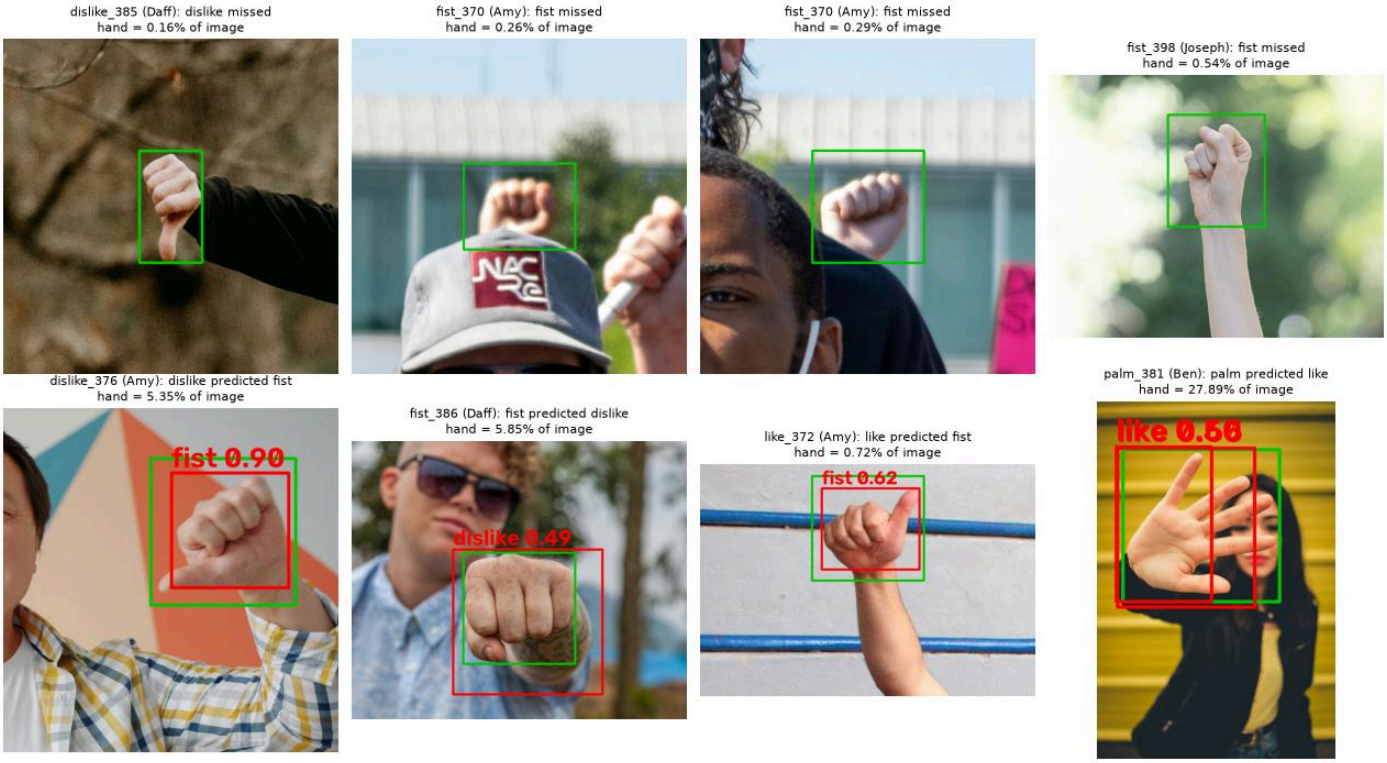




hand size	hands	recall	missed
small (<1%)	16	0.56	6
medium (1-5%)	86	0.84	13
large (>5%)	52	0.83	5

member	hands	predictions	correct	precision	recall	mean IoU
Amy	46	39	32	0.82	0.70	0.78
Ben	36	35	30	0.86	0.83	0.76
Daff	32	35	26	0.74	0.81	0.81
Joseph	40	39	36	0.92	0.90	0.72

Hand size is the clearest driver. The median labelled hand covers 3.5% of the image, but 16 hands cover under 1%, and only 56% of those are found. At imgsz 640, a hand covering 0.3% of the image is only about 35 px across after resizing, which leaves little detail. Size alone is not the cause, though: 386 of the 1280 training hands (30%) are also under 1% of the image, and the median training hand (1.6%) is smaller than ours. The difference is context. HaGRID's small hands belong to one person posing in a plain room; ours are in crowds, outdoors and at odd angles. **Per member**, the differences follow photo choice and labelling style. Amy's photos hold 10 of the 16 small hands, which lowers recall there. Joseph's photos have the lowest mean box IoU (0.72) even though its precision is 0.92: the model finds those hands, but the boxes are drawn differently, a labelling-style difference rather than harder photos.



Green = our label, red = prediction (conf ≥ 0.25). Top row: smallest missed hands. Bottom row: wrong-class predictions: fist/dislike and palm/like confusions.

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7. Speed and size

Per image at imgsz 640 (batch 1, RTX 5060 Ti): preprocess 2.1 ms, inference 12.1 ms, postprocess 2.4 ms, **16.6 ms in total (≈60 frames/s)**; repeated runs vary by a few ms. The model is 23 MB (11.1 M parameters, 28.4 GFLOPs). A webcam photo booth needs about 15-30 frames/s, so a GPU has headroom to spare. CPU-only speed was not measured.

8. Limits of this evaluation

- **Small test set.** 154 hands in 128 photos. The bootstrap interval for F1 in Section 3 spans 0.12.
- **Duplicate photos.** One pair of our test photos is the same picture contributed by two members (`dislike_388.jpg` (Daff) = `dislike_393.jpg` (Joseph)). Those hands count twice, so the test set has 127 distinct photos, not 128.
- **Labelling errors (human error).** Checking every photo against the model's predictions found label mistakes that were left in the test set: boxes drawn much larger than the fist in `fist_380`, `fist_381`, `fist_384`; gesturing hands that were never labelled in `fist_376` (2 raised fists); `fist_382` (2 raised fists in the crowd); `palm_388` (palms of the people behind the main subject); and an unclear gesture in `palm_376` (hand held across the face, back of the hand to the camera). A tight, correct box on a loose label overlaps it by less than 0.5 IoU and counts as a miss plus a false positive, and a detection of an unlabelled hand counts as a false positive. At conf 0.25 these cause 3 of the 24 missed hands and 4 of the 18 background predictions, so the scores on our photos (fist most of all) are slightly pessimistic.
- **Validation is optimistic.** It is HaGRID-only and drawn from the same distribution as training, so checkpoint selection does not reward robustness to our kind of photo.
- **One training seed.** Run-to-run variation was not measured; it should be kept in mind when comparing tuned models.
- **Labels differ by annotator.** Box styles vary between members, which depresses mAP50-95 and IoU-based matching.

9. Takeaways for hyperparameter tuning

1. **Small hands** are the main failure mode. We tested other image sizes (`Gesture_YOLO_Tuning_imgsz.ipynb`): team mAP50 was 0.830 at 480, 0.845 at 640, 0.848 at 800, 0.828 at 960, and small hands found were 8, 9, 10, 9 of 16, so no size changed small-hand recall by more than 1 hand. The training set is not short of small hands; it is short of our kind of scene (crowds, outdoors, several people). More varied training photos and context-changing augmentation (mosaic, mixup, scale) are the more promising levers.
2. **Fist vs dislike** needs more varied fist examples or more epochs of fine-grained learning; watch fist recall in every comparison.
3. Judge tuned models on **our photos** (team-only mAP50 0.845, F1 0.821 at conf 0.25), not on validation, which is saturated at 0.99.

PART B · HYPERPARAMETER TUNING

Image size (`imgsz`) compared with the baseline

Tuned by Ben Nichiporik · YOLOv8s gesture detector · imgsz 480 / 800 / 960 vs baseline 640 · November 2026

0.848
best team mAP50 (imgsz 800)
vs 0.845 baseline: +0.003

±0.017
largest change in team mAP50
(single-model noise ±0.06)

8 / 9 / 10 / 9
small hands found of 16
at imgsz 480 / 640 / 800 / 960

0.821
baseline F1 at conf 0.25,
best of all four runs

Result: changing the image size did not improve the baseline. Team-only mAP50 moved by at most 0.017, well inside the ±0.06 uncertainty of a single model on our 128 photos. imgsz 800 is best on paper, but the baseline (640) keeps the best F1 at the deployment threshold and the best speed/accuracy balance. A larger input did not help small hands, because the baseline's problem is unfamiliar *scenes*, not too few pixels.

B1. Setup and hypothesis

`imgsz` is the size (longest side, in pixels) that every image is resized to for training and inference. The baseline evaluation (Part A) showed that the model's main weakness on our own photos is **small hands**: only 56% of the 16 hands covering under 1% of the image were found. At imgsz 640 those hands are only about 25-62 px across (median hand 119 px).

Hypothesis. A larger `imgsz` (800, 960) keeps more pixels per hand and should raise recall on small hands, especially fists, at the cost of slower training and inference. A smaller one (480) shows how much accuracy we lose for speed, which matters for a real-time photo booth.

run	imgsz	epochs run	best epoch	train min
imgsz_480	480	50	41	5.3
baseline	640	50	41	8.5*
imgsz_800	800	50	38	17.0
imgsz_960	960	41	26	26.0

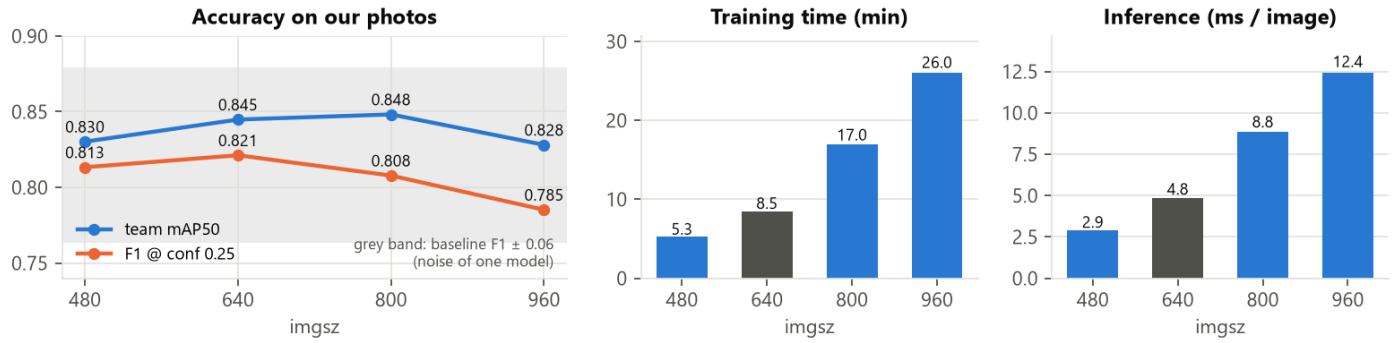
Only `imgsz` changes. Everything else is the baseline: YOLOv8s (COCO), SGD lr0 0.01, 50 epochs, patience 15, batch 16, seed 0, deterministic. *8.5 min on an idle GPU (15.3 min in the final re-run on a shared GPU, identical model).

Fair comparison. Same split, seed and evaluation code (`gesture_eval.py`). Each model is scored at the size it was trained at, and the headline is our own photos only: 128 images, 154 hands, at confidence ≥ 0.25 and IoU ≥ 0.5.

B2. Results compared with the baseline

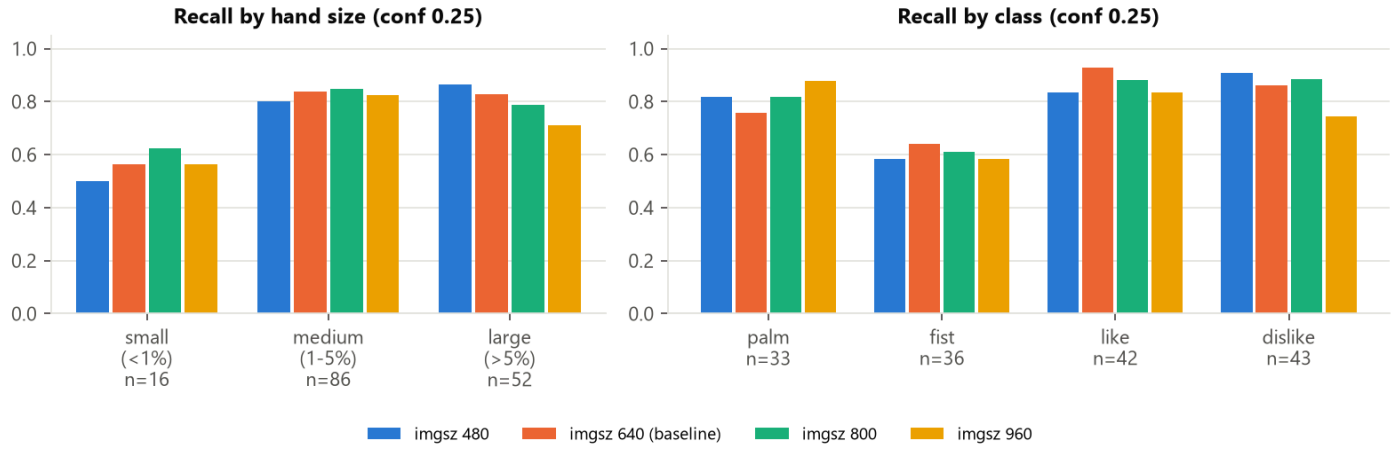
imgsz	val mAP50	team mAP50	team mAP50-95	fist mAP50	precision	recall	F1 @ 0.25	Δ team mAP50	Δ F1 @ 0.25	small hands	fixed / broken
480	0.995	0.830	0.426	0.618	0.855	0.780	0.813	−0.015	−0.008	8 / 16	9 / 11
640 (baseline)	0.990	0.845	0.418	0.625	0.894	0.775	0.821	—	—	9 / 16	—
800	0.994	0.848	0.429	0.642	0.894	0.765	0.808	+0.003	−0.013	10 / 16	9 / 9
960	0.995	0.828	0.426	0.594	0.880	0.751	0.785	−0.017	−0.036	9 / 16	12 / 19

mAP, precision and recall from Ultralytics on our photos (val = HaGRID validation). F1, small hands (<1% of the image) and fixed / broken (hands the baseline got wrong that this run gets right / the reverse) at conf 0.25, IoU 0.5.



Grey band: baseline F1 ± 0.06, the 95% bootstrap half-width of one model's F1 on our photos. Inference is per image at batch 16 on an RTX 5060 Ti and varies by 1-2 ms between runs.

- **No run beats the baseline by more than noise.** Validation is saturated (0.990-0.995) for every size, so it can't separate the runs either.
- **imgsz 800 is best on paper, but only just**: highest team mAP50 (0.848, +0.003), mAP50-95 (0.429) and fist mAP50 (0.642, +0.018). Its precision is the same as the baseline's and its recall is lower (0.765 vs 0.775), so F1 at the deployment threshold is slightly *worse* (0.808 vs 0.821). It fixed 9 hands and broke 9.
- **imgsz 960 is the worst for its cost.** It early-stopped at epoch 41 (best epoch 26), took 3× the baseline's training time, fixed 12 hands but broke 19, and large-hand recall fell from 0.83 to 0.71.
- **imgsz 480** is the fastest and only slightly worse (−0.015 mAP50), but it finds the fewest small hands (8 / 16).



Small hands barely moved. 8/16 at 480, 9/16 at 640, 10/16 at 800, 9/16 at 960. One extra hand at 800 is not a real improvement on a sample this small, so a larger `imgsz` did not fix the baseline's main weakness. Per class, fist stays the weakest class at every size (recall 0.58, 0.64, 0.61, 0.58).

B3. Why a larger image size didn't help

Resolution was not the real problem. The training set already has plenty of small hands: 386 of the 1,280 HaGRID training hands (30%) cover under 1% of the image, and the median training hand (1.6% of the image) is *smaller* than ours (3.5%). The gap is **scene context**. HaGRID shows one person standing and facing a phone camera, mostly in a plain indoor room. Our missed hands are in crowds, outdoors, at odd angles and against busy backgrounds. A larger input gives the model more pixels, but it doesn't teach it what those scenes look like.

imgsz 960 may be under-trained. Its best checkpoint is from epoch 26, so it effectively trained for about half as long as the others. That probably explains part of its drop on large hands.

imgsz	baseline weights run at this size	model trained at this size
480	0.837	0.830
640 (baseline)	0.845	0.845
800	0.823	0.848
960	0.785	0.828

Control: team mAP50 of the baseline weights (trained at 640) run at each size, compared with a model trained at that size.

The control fits this explanation. Running the 640-trained baseline at a larger size *lowers* team mAP50 (0.845 → 0.823 at 800 → 0.785 at 960). Our hands are already larger on average than HaGRID's, and upscaling pushes them further from the sizes seen in training. Training at the larger size only wins back what the mismatch cost.

Label errors. The labelling mistakes listed in Part A, Section 8 are still in the test set. Every run is scored against the same labels, so the comparison between sizes is still fair, but absolute scores (fist especially) are slightly pessimistic.

B4. Choice and next steps

1. **For the group comparison, my best value is imgsz 800** (best team mAP50, mAP50-95 and fist mAP50). The gain over the baseline is +0.003 mAP50, effectively a tie, and it costs about 2× the training time and 1.8× the inference time.
2. **For the photo booth, keep imgsz 640.** It has the best F1 at the deployment threshold (0.821) and the best balance of speed and accuracy.
3. **Fix the data, not the resolution.** More training photos with varied scenes (crowds, outdoors, several people) and augmentation that changes the scene (`mosaic`, `mixup`, `scale`, colour changes) target the real weakness. If another member's hyperparameter improves scale robustness (e.g. `scale` or `mosaic`), combining it with imgsz 800 is the combination worth testing.